CHAPTER THREE

Climate Change Impact and Adaptation for Urban Drainage Systems

Yekenalem Abebe, Solomon Tesfamariam
University of British Columbia, School of Engineering, Kelowna, BC, Canada

3.1 Introduction

IPCC's Fifth Assessment Report concluded that, due to global warming, heavy precipitation events are expected to increase in frequency and intensity under climate change (Pachauri et al., 2015). In Canada, annual mean temperatures have increased by 1.6°C over the last 50 years (Environment and Climate Change Canada, 2016). Similarly, annual precipitation has increased from 5% to 35% in southern Canada over the last century (Groisman and Easterling, 1993). Changes in precipitation patterns towards more intense storms could lead to widespread pluvial flooding, aggravated by inappropriate land-use planning, increased paving, and loss of water storage space. Drainage networks that are designed based on historical climate regimes may be defunct in the future. Pluvial flooding occurs when extreme rainfall overwhelms drainage systems and the excess runoff cannot be absorbed (Maksimović et al., 2009). This could result in significant damage costs and environmental impact. The damage cost could further increase due to the synergetic effects of infrastructure ageing, climate change, and urbanisation. In 2016, more than $10 billion worth of stormwater assets was in poor or very poor condition across Canada, and with the current reinvestment level, a further decline in asset condition is expected (Canadian Infrastructure Report Card, 2016). On the other hand, a growing urban population requires new infrastructure development and alteration in land cover towards predominantly impervious surfaces. Such shifts will reduce the amount of water that infiltrates and it can intensify flood hazards (Harbor, 1994; Pauleit et al., 2005). Furthermore, impervious areas collect and accumulate pollutants and particles that are washed off during rainfall and pollutes receiving water bodies.

Climate Adaptation Engineering
https://doi.org/10.1016/B978-0-12-816782-3.00003-6

Restoring flooding and associated damages is expensive for property owners, municipalities, and insurance companies. Hence, there is a critical need to support municipalities in their decision-making processes to effectively manage stormwater infrastructure, prioritise investment decisions, and optimise cost while incorporating climate adaptation strategies in the process. To date, there have been various reported studies on climate change impact and adaptation of urban drainage systems. Waters et al. (2003) investigated the impact of a 15% increase in extreme rainfalls on a typical urban catchment in southern Ontario and proposed adaptation alternatives. Kirshen et al. (2014) presented a case study on Somerville's urban drainage system in Massachusetts. They used multicriteria scenario analysis to evaluate alternative adaptation strategies, based on the volume of flooding, volume of combined sewage discharged, and peak flow rate. Zhou et al. (2012) investigated three climate adaptation options in Odense, Denmark and presented a risk-based framework to assess the economic feasibilities of each option. Kang et al. (2016) studied alternatives to improve the drainage system in Incheon, Korea and proposed a seven-step decision-making framework. Berggren et al. (2011) investigated the hydraulic performance of urban drainage systems in future climate condition in Kalmar, Sweden. Semadeni-Davies et al. (2008) assessed the potential impacts of climate change on stormwater flows in Helsingborg, South Sweden. They concluded that urbanisation and projected increases in heavy rainfalls are set to increase flood risk and recommended the installation of sustainable urban drainage systems (SUDS) techniques to relieve the adverse impacts. Most of the reported studies provide detailed investigation on a selected section of a drainage network.

In this chapter, we present a three-phase framework that accounts for city-scale flood vulnerability assessment as well as detailed mitigation investigation. Furthermore, important steps in performance, hazard, and risk assessment are outlined. In addition, economic analysis of adaptation options as a way of selecting the most appropriate option is discussed. The proposed framework is described in detail in Section 3.2. In Section 3.3, the framework is applied on a case study area in Vernon, Canada.

3.2 Methodology

The proposed methodology outlines a comprehensive framework to assess climate change impact and plan adaptation alternatives for drainage infrastructure (Fig. 3.1). It follows a top-down approach that accounts for large-scale vulnerability assessment as well as detailed adaptation investigation. In addition,

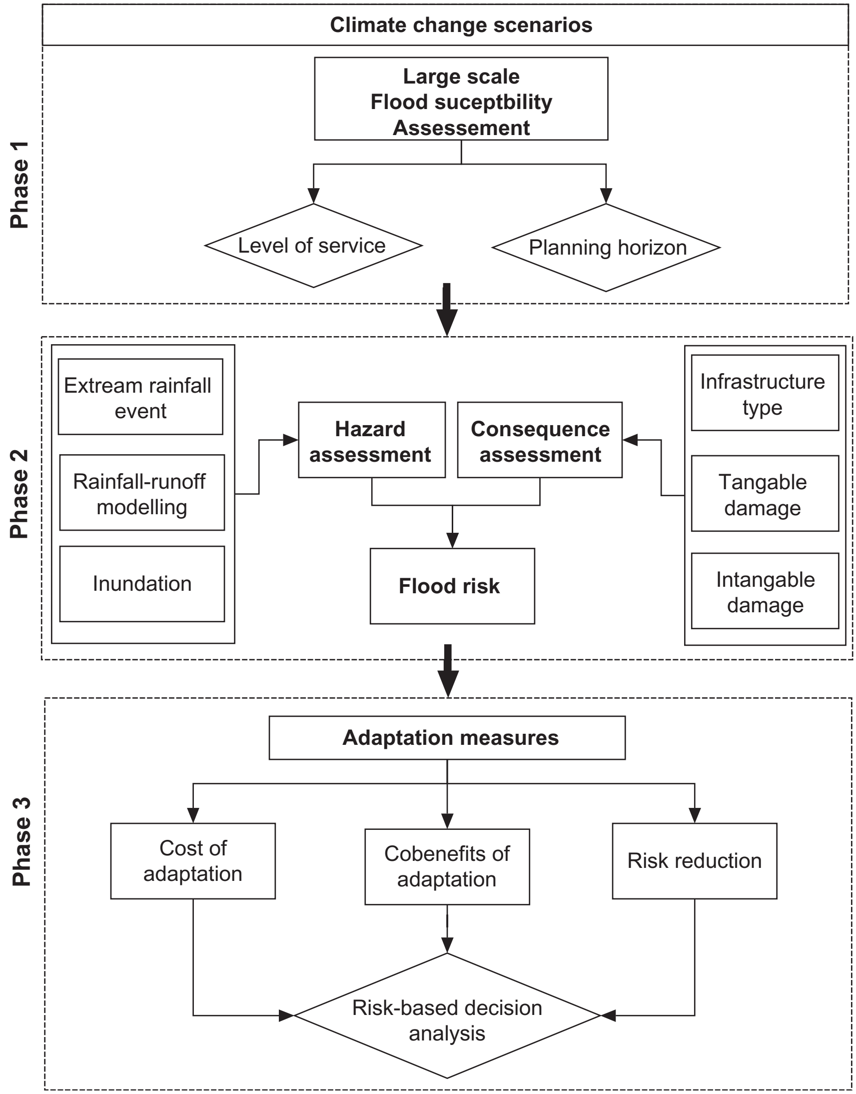

Fig. 3.1 Climate change impact and adaptation decision-making framework for urban drainage systems.

important steps in performance, hazard, and risk assessment are outlined and economic analysis of adaptation alternatives is discussed. The framework is subdivided into three phases:

- Phase 1: city-scale pluvial flood vulnerability assessment is carried out to identifying critical locations for further investigation. This step is crucial to focus resources and carry out detail studies only on critical sections of the drainage network. Furthermore, the planning horizon and the required level of service (LOS) need to be defined before proceeding to the next phase.

- Phase 2: detailed climate change impact assessment is carried out. The performance of the drainage system for current and future climate conditions is assessed using one-dimensional-two-dimensional (1D-2D) inundation simulation.
- Phase 3: adaptation measures relevant to urban drainage systems are identified and their respective performance in terms of risk reduction is assessed. Finally, the overall benefit of adaptation options is compared to choose among potential alternatives.

3.2.1 Phase One: City-Scale Vulnerability Assessment

In spite of increasing attention towards sustainable practices for stormwater management, adaption planning to pluvial flooding remains a challenging task due to the complexity of urban environment (Zhou et al., 2012). Large-scale modelling of an urban drainage network is difficult and time-consuming. Urban drainage systems consist of several underground structures (such as pipes and manholes) and overland flow routes (including roadside ditches, roadways, storage ponds, and swells). These interconnected structures often exhibit complex nonlinear dynamics, which makes it difficult to optimise the technical performance of the system and estimate the net benefits of adaptation measures. In addition, urban areas have a complex socioeconomic interaction that makes it difficult to quantify damage cost and risk that is required for adaptation planning.

In the first step of the framework, large-scale flood vulnerability assessment is proposed to identify critical locations that might be highly affected by climate change. This is crucial for municipalities to focus resources and carry out detailed investigations only on the critical neighbourhood. Thus, a flood vulnerability assessment approaches using a coupled GIS and Bayesian belief network (BBN) model that can capture the casual nexus between pluvial flood causative factors is proposed (Fig. 3.2). Details on the methodology and implementation are reported in Abebe et al. (2018). To construct the BBN model, important factors that can contribute to the occurrence of urban flooding are identified. The factors used are land cover, population density, slope, soil drainage class, drainage density, DEM, rainfall, and drainage capacity. The number of reported basement flooding and sewer backup damages can be used to represent the flood vulnerability index (FVI). At the

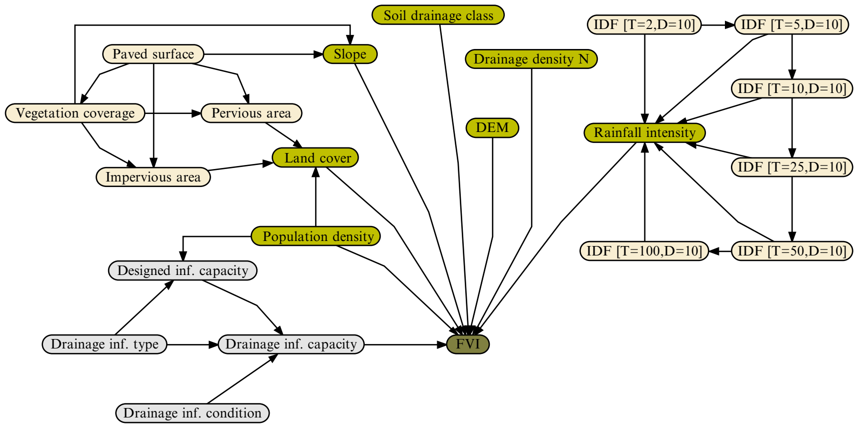

Fig. 3.2 BBN-based pluvial flood vulnerability assessment model (Abebe et al., 2018).

end of this phase, critical locations with high FVI will be identified for further analysis.

3.2.1.1 Level of Service

The required level of flood protection and water quality standards are the main considerations in urban drainage systems design. Stormwater drainage systems are composed of minor and major systems. The minor drainage system is designed to handle storms that are more frequent. Paths taken by runoff beyond the capacity of the minor system are called the major system. The major system is intended for the less frequent storm. It consists of natural waterways, temporary overload relief swales, and green areas. The major and minor systems are closely interrelated and their design and management need to be done in conjunction to achieve an overall stormwater management objective (Maksimović et al., 2009).

The performance of a drainage system can be measured by its hydraulic capacity to safely convey runoff through the catchment without causing flooding. Municipalities usually set a design standard based on the return period that states the required level of protection. For example, the city of Vernon, Canada recommends that the minor system should be capable to handle a five-year storm in low-density residential areas, while, the major system is designed to handle a 100-year storm (Bailey, 1992). For the storm sewer and the major overland flow path, this corresponds to the annual probability of having an overflow is 0.20 and 0.01, respectively. Therefore, the drainage system LOS can be defined by the precipitation return period where flood damage starts (Arnbjerg-Nielsen and Fleischer, 2009).

The ability of urban drainage systems to function adequately in a variety of conditions is an important system characteristic (Nanos and Filion, 2016). Climate change induced upsurge in extreme rainfall events can affect the performance of stormwater systems. To date, the design of stormwater infrastructure is based on an underlying assumption that the probability distribution of precipitation extremes is statistically stationary (Rosenberg et al., 2010). However, this assumption is invalid due to climate change (Srivastav et al., 2014). Therefore, the intensity and frequency of pluvial flood hazard can significantly increase in the future.

3.2.2 Phase Two: Climate Change Impact Assessment

Stormwater drainage infrastructure are planned based on local weather conditions. However, hydrologic data are no longer static over time due to climate change and a new design approach is required. The combination of climate change, land-use change, and infrastructure deterioration could exacerbate urban flooding. When municipalities look to improve existing infrastructure and build new stormwater systems, these changing conditions have to be considered in the planning process. In this section important steps in drainage systems performance, flood risk, and climate change impact assessment are discussed.

3.2.2.1 Hazard Assessment

Urban pluvial flooding process is a function of precipitation, runoff generation rate, overland, and sewer flow capacity of an area. Each of these hydrological and hydraulic processes must be analysed to assess flood hazard. This requires rainfall-runoff modelling, sewer flow modelling (1D simulation), and overland flow routing (1D or 2D simulation).

Extreme Rainfall Events

The first important step in hazard assessment is to determine the system loading or extreme rainfall events. Intensity-duration-frequency (IDF) curves are most commonly used to characterise precipitation events. The IDF curves provide precipitation accumulation depths for various return periods (T) and different durations (D). They are typically developed by fitting extreme value distributions, like Gumbel, generalised extreme value (GEV), or log-normal, to annual maximum precipitation (AMP) time series (Mirhosseini et al., 2013). However, IDF curves prepared based on historical data must be updated to reflect the expected shift in the future climate.

General circulation models (GCMs) and regional climate models (RCMs) provide the dynamics of the climate system and projections of future climate parameters (Eden et al., 2014). These models have low spatial resolution and do not provide the relevant temporal and spatial details required for drainage systems planning. Therefore, downscaling techniques (dynamic or statistical) are employed to prepare localised and high-resolution data. Statistical procedures generally formulate predictor-predicted relationships between GCMs and locally observed data (Herath et al. 2016). Various statistical downscaling procedures have been proposed for updating IDF curves (e.g. Hanssen-Bauer et al., 2005; Jacobeit et al., 2014; Nguyen et al., 2007; Wilby et al., 1998). In this study, equidistance quantile matching (EQM) method proposed by Srivastav et al. (2014) is used. The first step in EQM method is to formulate a statistical relationship between the observed station and the GCM historical (baseline) AMPs. In the second step, it establishes a statistical relationship between the GCM base period and the future projections. Finally, the relationship between the first two steps is used to update the IDF curves for future periods. More details on the methodology can be found in Sandink et al. (2017) and Srivastav et al. (2014).

Rainfall-Runoff Modelling

To quantify flood hazard, first the rainfall-runoff relationship must be established. This is a complex process and many models are available that vary in their accuracy (Lee et al., 2005; Rajurkar et al., 2004). The time-area method is one of the simplest and fastest alternative to estimate total runoff flow to manholes (O'Loughlin et al., 2010). This method is based on a rainfall excess hyetograph and time-area representation of the contributing catchment. Catchments are defined by the percentage of impervious area, time of concentration, and the time-area curve. The excess runoff depends on the initial losses from wetting and depression storage and continuous hydrological losses caused by evapotranspiration and interception.

Inundation Modelling

Inundation modelling can be based on physical-based models resolving continuity and momentum water equations or conceptual models that satisfy only continuity equation and use theoretical relationships among parameters (Ochoa-Rodríguez et al., 2013). Since physical-based models are computationally intensive and require longer processing time, conceptual models using GIS-based surface flow models and digital elevation model (DEM) are an alternative and simplified approach. However, this approach might affect forecast precision and it cannot predict water velocity (Liu and Pender, 2010). Examples of such methods include Rapid Flood Spreading Model (Lhomme et al., 2008; Wang et al., 2010), and Cellular Automata approaches (Guidolin et al., 2016). Van Dijk et al. (2014) compared these techniques with physical-based models and found that surface flow models prove to be an easy and fast alternative to identify vulnerable locations in hilly and flat areas. Whereas coupled 1D-2D simulation models give better results in areas at the transition between hill and flat. Pluvial inundation modelling techniques are evolving towards coupled 1D-2D simulation models, which can simulate the interaction between surface and sewer flows (Van Dijk et al., 2014). In this study, a coupled 1D-2D simulation is used to analyse the inundation extent for different climate change projection scenarios.

3.2.2.2 Consequence (Damage) Assessment

Flooding can have physical, social, or ecologic impacts (Merz et al., 2010). Tangible damages are quantified in monetary terms, whereas, intangible damages like social and ecological impact are difficult to estimate (Natural Resources Canada, 2017). Social impact assessment focuses on victims' psychological and emotional ordeal, while, ecological impact assesses the disruption to the natural ecosystem (Kubal et al., 2009). Methods for quantifying intangible damages are not well developed and most urban flood risk analysis studies consider only tangible damages (Lekuthai and Vongvisessomjai, 2001). Tangible damages can be direct damages that occur immediately or indirect damages that occur over time and can impact activity in surrounding areas (Thieken et al., 2009). Usually, indirect damages are estimated as a percentage of the direct damage estimation. Direct damages can be further grouped into structural, internal, or external (Natural Resources Canada, 2017). These damages can be assessed using depth damage, fragility, or loss curves, which describe the relationship between the depth of inundation and the economic loss. Since, it is impractical to prepare individual curves for all structures in a study area, a classification scheme that groups similarly behaving structures into one class is typically applied. These relationships can be established from historical flood events and damage data or can be synthesised from a detailed loss assessment on a sample structure. Flood damage is influenced by multiple factors, such as flow velocity, duration, contamination, etc., therefore, loss functions based only on inundation depth may have a large uncertainty (Thieken et al., 2009).

3.2.2.3 Flood Risk

Flood risk in a given planning horizon is determined by aggregating the damage costs resulting from various extreme rain events, based on their probability of occurrence (Olsen et al., 2015). High probability events have a lower damage cost, on the other hand, events that cause substantial damage may not occur frequently (see Fig. 3.3A). It is important to consider multiple simulations to cover a large range of return periods. Combination of these events with their corresponding return periods yields the risk-density curve $D(p)$ (Arnbjerg-Nielsen and Fleischer, 2009) (see Fig. 3.3B). The expected annual damage $E(L)$ can be computed by integrating the risk-density curve over all probabilities (p) and catchment area (A) as:

$$E(L) = \iint D(p)\,dp\,dA \tag{3.1}$$

where $D(p)$ is the damage for a given return period ($T = 1/p$). The integral is solved in the range between zero and infinite, $T \in [0, \infty]$. However, it is not practical to simulate all possible return periods. Consequently, various approximation techniques can be used to estimate $E(L)$ (Olsen et al., 2015). Numerical integration over all probabilities is one option, which requires the calculation of the associated risk for n relevant return periods. It is also possible to find analytical solution by assuming a log-linear relationship between damage cost and return period T. Simulated time series is another method to estimates the average expected damage costs for a given observation period (Olsen et al., 2015). Applying a log-linear relationship between damage cost and return period, it is possible to calculate $E(L)$ with only a few hazard simulations (Arnbjerg-Nielsen and Fleischer, 2009). In this study, the simulated time series method is used. This method estimates

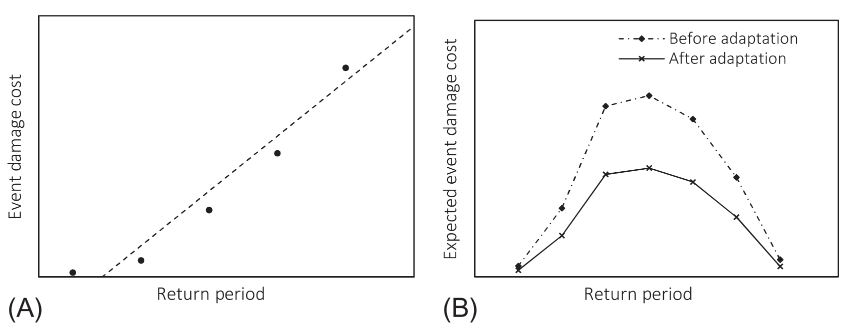

Fig. 3.3 Flood risk assessment (A) return period and damage cost relationship and (B) risk-density curve.

Table 3.1 Expected Annual Damage Calculation

Rank	Return Period	Damage Cost for Event
1	T_1	C_1
2	T_2	C_2
…	..	..
M	T_m	C_m
		$E(L) = \sum_{i=1}^{m} \frac{C_i}{n}$

the number of exceedance events with in a given period and aggregates the cost (Table 3.1). The rank and return periods of the events can be determined by using the median plotting position using Eq. (3.2):

$$T = \frac{1}{p} = \frac{n + 0.4}{m - 0.3} \tag{3.2}$$

where n is the duration of the simulated time series, T and p are the return period and the exceedance probability, respectively, m is the rank of the event. For example, when simulating a total study period of 100 years, and assuming the LOS for the stormwater drainage system to be $T = 5$ years; there will be 18 flood events with the first ranked event having 129.19 return period and the last ranked event with 5.11 return period. Assuming a log-linear relationship between flood damage cost and return period all the flood events damage cost can be estimated as a multiple of the 100-year flood event. The 100-year and 5-year return period events are considered to have 100% and 0% damage cost, respectively. More details on the methodology can be found in Olsen et al. (2015) and Arnbjerg-Nielsen and Fleischer (2009).

3.2.3 Phase Three: Climate Change Adaptation

Although municipalities are facing severe and multifaceted challenges, several options can help to minimise climate change impacts. Climate change adaptation involves a variety of measures to reduce risk. The area between the risk-density curve before and after the adaptation measures represents the risk reduction due to the adaptation measures (Fig. 3.3B). This can be achieved by managing hazard to minimise the frequency and intensity of flood events or by improving the resiliency of vulnerable structures. The appropriate approach depends on the specific context of the study area. In general, adaptation strategies can be understood in four dimensions based on the intent, timing, spatial, and temporal scope (Fig. 3.4). The intent of the

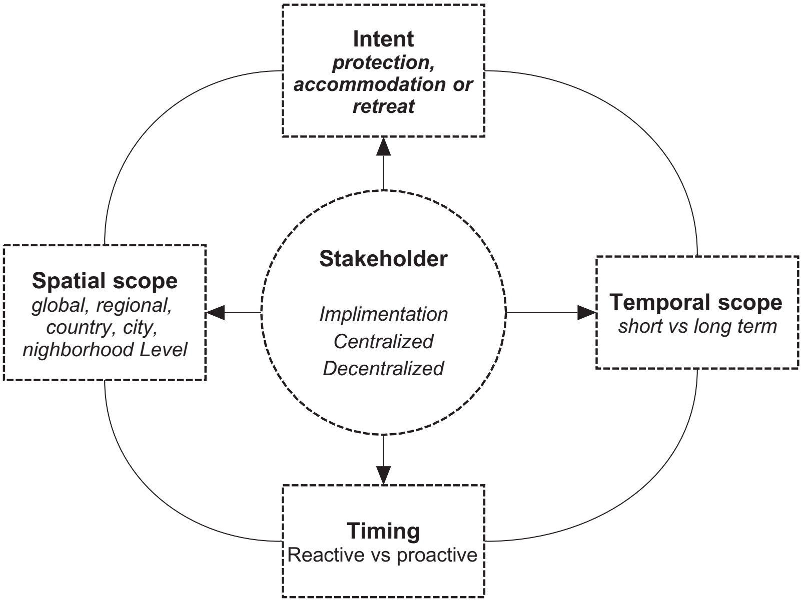

Fig. 3.4 Four dimensions of adaptation strategies.

adaptation could be protection, accommodation, or retreat (Zhou et al., 2011). Protection referrers to the construction of a barrier lessen the impacts of climate change, such as a seawall to protect against coastal flooding. Accommodation allows the flooding to occur in a managed manner. Examples of accommodation actions could include developing evacuation plans, building temporary shelters, and purchasing insurance. The last option is to retreat or move away from the hazard, for example, relocating from flood-prone areas.

Furthermore, climate change adaptation can be reactive or proactive (anticipatory) according to the timing of the action (Smit et al., 1999). Proactive adaptations are implemented before the impact is observed while reactive adaptation actions take place after. Reactive adaptation is often needed to address current impacts, while proactive adaptation is planned for a comprehensive long-term response. In most cases, proactive adaptations are more cost-effective than reactive adaptations. In addition, the scope of adaptation plans widely varies in temporal and spatial scale. Adaptation actions can be large scale or focused on a specific location. It is often desired to have adaptation actions with longer and wider effects. Furthermore, adaptation strategies should be flexible and resilient in terms of both temporal and spatial scope.

Available adaptation measures for urban drainage systems can be discussed under three groups: source control, system control, and downstream measures (Zhou, 2014). Source control helps minimise excess runoff upstream. This can be achieved by facilitating local infiltrations and storage systems. Typical examples include permeable pavements, green roofs, blue roofs, ponds, dry wells, and so on. Downstream control measures maximise the conveyance capacity and efficiency of the drainage system. Examples include pipe enlargement, detention ponds, or relief channels. System controls aim to reduce hazard impact on vulnerable structures, such measures can include topographic or land-use changes and flood defence structures. These adaptation options can be hard (structural) or soft (non-structural) measures depending on their physical impact on the environment.

Cost-benefit analysis is an important tool to understand the relative economic advantages of different adaptation alternatives (Bastidas-Arteaga and Stewart, 2015; see also Chapter 1). It is a systematic process for calculating the net benefits of a decision, policy or a project. Net present value (NPV) is one of the most popular analysis metrics. It computes the difference between the present value of cash inflows and outflows. A positive net present value indicates that the projected generated investment exceeds the anticipated costs. In this case, the risk reduction due to adaptation measures is compared against the investment needed to implement the adaptation measures. An investment with a larger NPV will always be preferred and if NPV is negative, it will result in a net loss. NPV is calculated as follows:

$$\mathrm{NPV} = \sum_{t_o}^{t_e} \frac{E(L)\Delta R - C_{\mathrm{adapt}}}{(1+r)^t} \tag{3.3}$$

where $E(L)$ is the expected annual damage, ΔR is the risk reduction due to adaptation measures, C_{adapt} is the investment cost of adaptation measures, r is the discount rate, and t_o and t_e are the start and end time of the analysis period, respectively.

3.3 Case Study

The city of Vernon is found in the Thompson Okanagan region of British Columbia, Canada (Fig. 3.5). It has a total population of around 40,000 and an area of 95 km^2. The storm drainage pipe network contains 3979 mains and 7414 service lines. Development is mainly concentrated

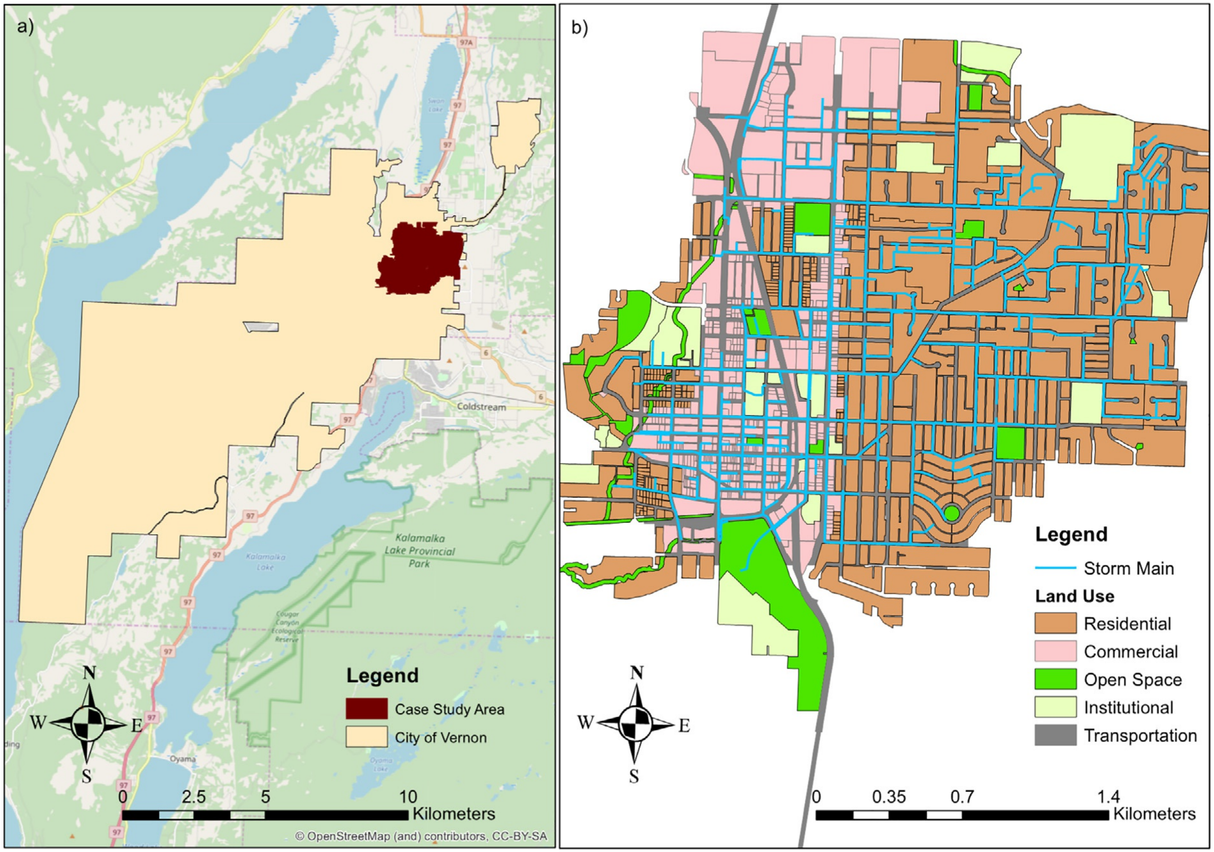

Fig. 3.5 Case study area (A) City of Vernon boundary and (B) close-up view of the case study area.

in the city centre and along the Okanagan lake coastlines. The city centre is located far from the lake and Vernon creek drains the city's storm network into the lake. For this case study, we have zoomed into the central part of the city, where a comprehensive storm network data is available (Fig. 3.5). Our study is limited to pluvial flooding in relation to stormwater drainage system capacity. The total catchment area considered in the study is 5.5 km^2. The catchment has mainly residential detached houses and commercial areas.

3.3.1 Stormwater Drainage System

Pipes, manholes, inlet, and outlet structures are the main components of storm sewer network. Pipes can be grouped into the trunk (main) and service pipes. A service line is smaller in diameter and is tributary to the trunk line. The study catchment contains 15 km of service pipes and 61 km of storm main. Manholes provide access to underground storm sewers for inspection and cleanout. They are provided where there is an abrupt change in direction, gradient, or pipe diameter and at major junctions where multiple storm drains converge. They can be made of either masonry or concrete, usually with cast iron or concrete covers. There are 836 manholes in the case study area. Storm drain inlets are used to collect urban runoff and discharge it to an underground sewer system. Inadequate attention to inlet capacity can cause an undue hazard to the security of many urban activities. There are three main types of inlet structures. Grate inlets consist of an opening in the gutter covered by screens. Curb-opening inlets are vertical openings in the curb covered by a top slab. Slotted drain inlets consist of a slotted opening with bars perpendicular to the opening. Slotted inlets function as weirs with flow entering from the side. Furthermore, there are 30 outfalls in the case study area that are located adjacent to Vernon Creek. Roadside gutters, ditches, and other overland open channel flow route are considered part of the major drainage system. Furthermore, stormwater best management practices (BMPs) are important components of a drainage system. BMPs provide water quality enhancement as well as improve the overall hydraulic capacity of the system. In the study area, there are 1545 catch basins and 9 drywells.

The performance of the stormwater drainage network is modelled using the commercially available software package MIKE FLOOD. MIKE FLOOD is a professional flood modelling application with 1D and 2D flood simulation engines. The core packages of the applications used in this study are MIKE URBAN for collection systems modelling and MIKE 21 for 2D surface flow. The sewer flow and overland flow are coupled at every manhole location. The drainage system performance and pluvial flood hazard are analysed for various climate change scenarios.

3.3.2 IDF Curves

Environment and Climate Change Canada (2017) prepares IDF curves for all weather stations across Canada based on historical rain records (Fig. 3.6). However, these curves need to be updated to reflect future changes in precipitation pattern. Pacific Climate Impact Consortium (PCIC) offers statistically downscaled daily Canada-wide climate scenarios, at a gridded resolution of roughly 10 km for the simulated period of 1950–2100 for three Representative Concentration Pathways–RCP2.6, RCP4.5, and RCP8.5 (PCIC 2013). Based on these downscaled models, existing IDF curves can be updated and used to investigate climate change impact on the study area's drainage system.

In this study, we used the web-based IDF updating tool for Canada developed by Simonovic et al. (2016). The new version of the tool utilised the spatially downscaled models developed by the PCIC and applied EQM method for temporal downscaling. They used GEV distribution to fit the AMP series and update the IDF curves. The updated IDF curves for the Vernon weather station are shown in Fig. 3.7. For example, a 2-year return period and 5-min rainfall intensity are 3.24, 3.46, 3.51, and 3.61 mm/h for historical, RCP2.6, RCP4.5, and RCP8.5 scenarios, respectively. This means that the rainfall intensity expected for the same return period will increase in the future. Similarly, the return period expected for a given rainfall intensity becomes shorter (Table 3.2). Interpolating between the historical and updated IDF curves, we can estimate that a 100-years storm event will occur every 43.8, 30.8, and 15.5 years for RCP2.6, RCP4.5, and RCP8.5 scenarios, respectively. This will have a significant impact on the intensity and frequency of pluvial flood hazard in the study area.

3.3.3 Inundation

Based on the IDF curves, a 1-h triangular hyetograph is constructed to represent the design storm for the different climate scenarios. The study catchment is subdivided into parcel polygons and the percentage of impervious area is estimated for each parcel. Each parcel is assumed to be connected to the nearest manhole in the network. The time of concentration for each parcel is calculated based on the shape of the subcatchments. Assuming a

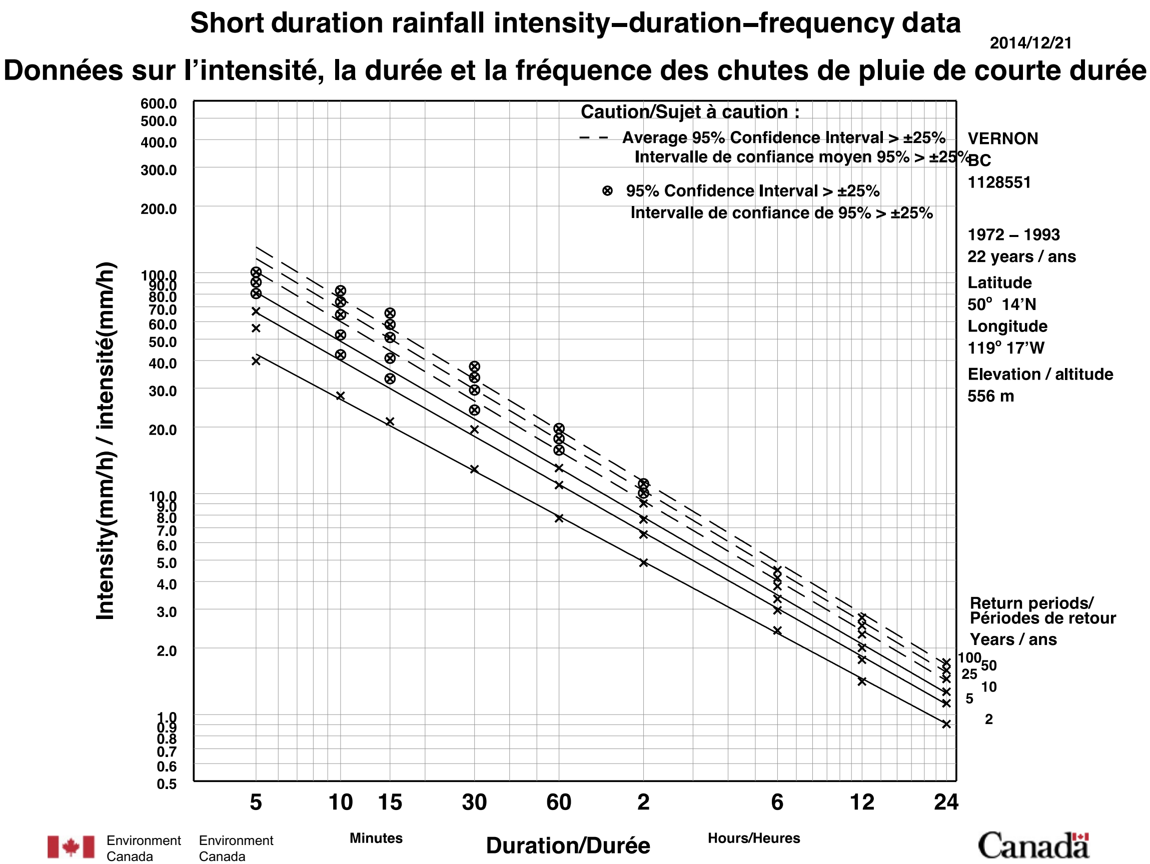

Fig. 3.6 IDF curves for Vernon, Canada (Environment and Climate Change Canada, 2017).

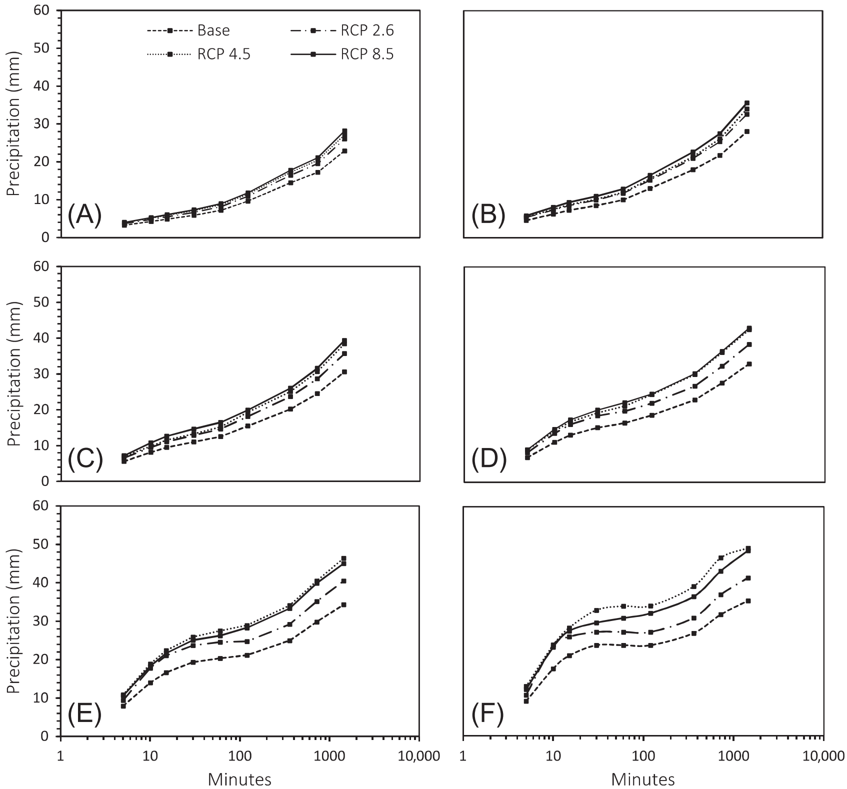

Fig. 3.7 Updated IDF curves for Vernon, Canada for return periods (A) 2 years, (B) 5 years, (C) 10 years, (D) 25 years, (E) 50 years, and (F) 100 years.

Table 3.2 Climate Change Impact on Return Period

Return Period	Today Base	Year 2096 RCP 2.5	RCP 4.5	RCP 8.5
T	100	43.8	30.8	15.5
LOS *T*	10	5.4	4.5	3.4

linear time-area curve representing the contributing part of the catchment surface as a function of time, the total runoff load to each manhole in the network is calculated. The overland flow from each overflowing manhole is routed using a 5-m resolution DEM and MIKE 21 Flood Screening Tool. Sample results for 50- and 100-year storms in base case scenario and for RCP 8.5 are plotted in Fig. 3.8. From Fig. 3.8, it can be seen that pluvial flood hazard will increase in future climate conditions.

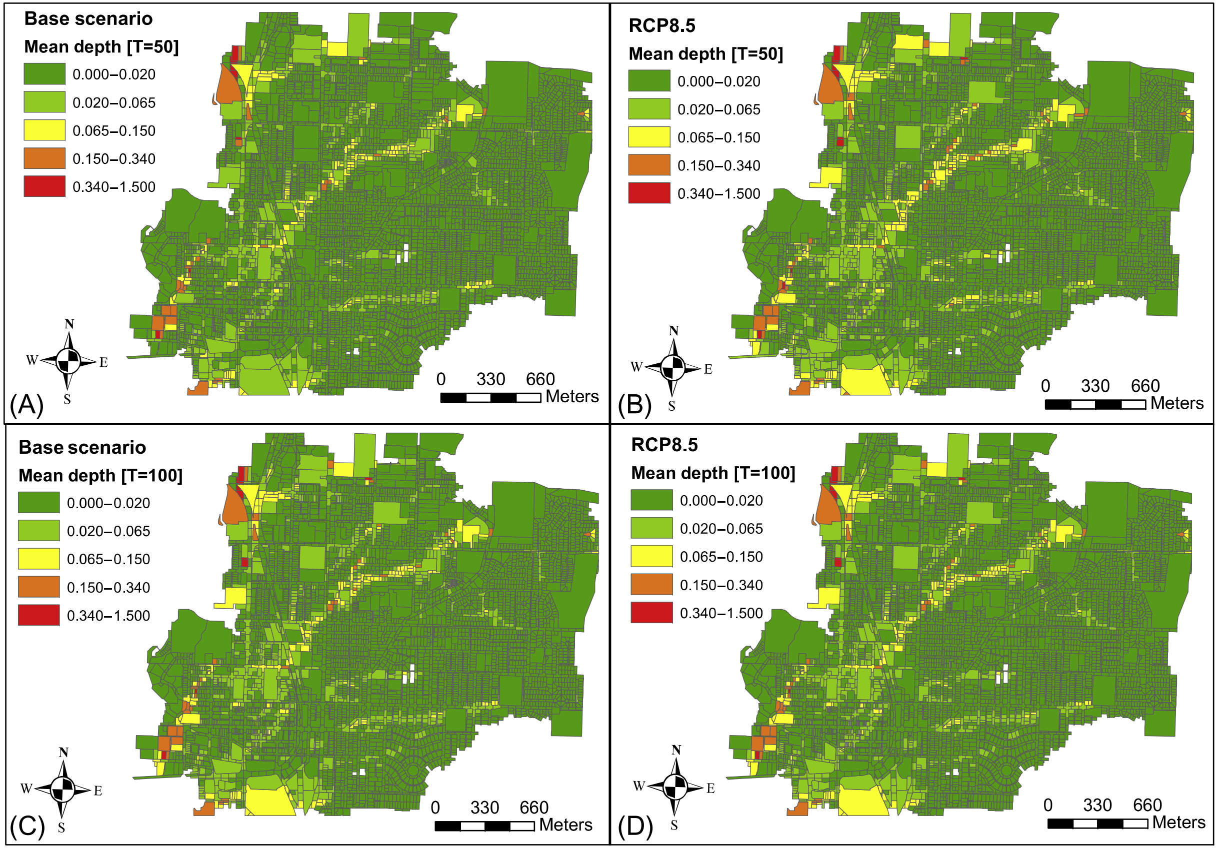

Fig. 3.8 Pluvial flood hazard for deferent scenarios (A) base scenario [$T=50$], (B) RCP 8.5 [$T=50$], (C) base scenario [$T=100$], and (D) risk-density curve.

Table 3.3 Expected Annual Damage Calculation for Base Case Scenario

Rank	*T*	% Cost	Expected Annual Damage *E*(*L*) [CAD]
1	129.1	111.1	$540,696
2	53.1	72.5	$353,167
3	33.4	52.4	$255,393
4	24.4	38.7	$188,801
5	19.2	28.40	$138,241
6	15.8	20.02	$97,471
7	13.4	13.00	$63,309
8	11.7	6.96	$33,908
9	10.3	1.66	$8,102

3.3.4 Cost of Climate Change

Assuming a 10-years LOS and planning horizon of 100 years, in the base case scenario, it is expected to have eight flood events in the next century, with the biggest event resulting in around $500,000 in damage cost and the minor event less than $10,000 (see Table 3.3).

Considering climate change, the number of flood event, as well as the damage cost, will increase significantly. The shift in return period based on the updated IDF curves is summarised in Table 3.2. Accordingly, the LOS will decrease in the future and more flooding events are expected. For example, considering RCP 2.5 scenario, there will be 17 flood events with the maximum damage cost of around $820,000. The event damage cost and expected total damage cost for the various climate scenarios considered are shown in Figs. 3.9 and 3.10.

3.3.5 Adaptation

Available adaptation methods can be discussed under the general concept of SUDS or more traditional sewer system expansion options, like pipe enlargement and underground storage tanks. In this study, we have considered two-adaptation options:

- *Option A*: it is assumed that individual households and commercial property owners will implement LID technologies to facilitate onsite infiltration and reduce peak runoff. LID technologies considered are drywells; green roofs, and permeable pavements. It is assumed that the total impervious area in the study area catchment will reduce by half.

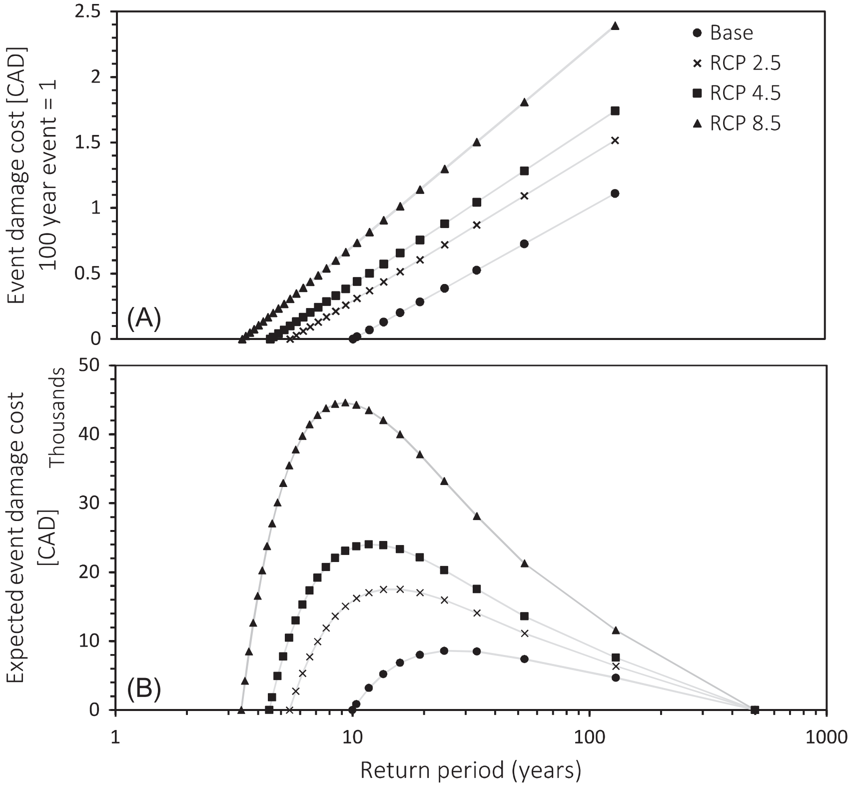

Fig. 3.9 Pluvial flood risk assessment (A) event damage cost and return periods, (B) risk-density curve.

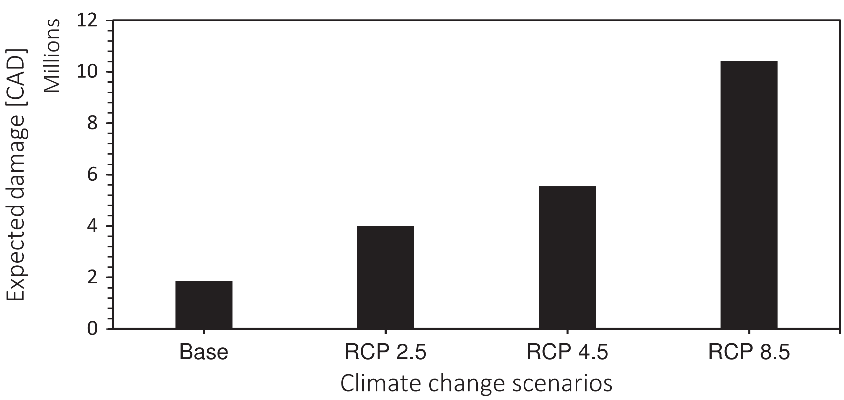

Fig. 3.10 Expected damage cost for different climate change projection scenarios.

- *Option B*: the municipality will construct underground storage tanks to relief the surcharge from critical nodes.

A NPV analysis is carried out for the two-adaptation options considered based on the RCP 4.5 scenario. For option A, the investment cost for each household and commercial site is estimated based on an average initial cost required to install green roofs, drywells, and permeable pavement. For option B, the municipality must upgrade the drainage system by installing storage tanks to relieve overflow runoff from critical manholes. Both adaptation options will help reduce the number of flood events, expected damage per event, and in general flood risk. For example, the damage cost for the first ranked flood event will reduce by 90% and 95% for adaptation options A and B, respectively. However, the NPV for option B is lower and option A provides the most benefit relative to the initial capital cost (Table 3.4). The event damage cost and expected annual damage for the various adaptation alternatives are shown in Fig. 3.11. The area between the risk-density curve before and after the two-adaptation alternatives represents the risk reduction due to the adaptation measures (Fig. 3.11B).

Table 3.4 NPV for Adaptation Measures

		NPV, in Million [CAD]	
Adaptation Options	**Investment Cost, in Million [CAD]**	**Discount Rate (1%)**	**Discount Rate (0.75%)**
(A) Infiltration	$51	$3.0	$19.4
(B) Storage tank	$60	−$8.2	$7.3

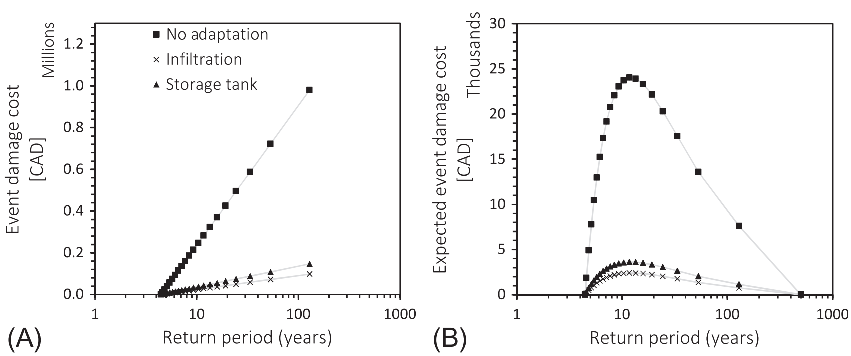

Fig. 3.11 Adaptation alternatives (A) event damage cost and return periods (B) risk-density curve.

3.4 Summary

There is a consensus regarding climate change and its impact on precipitation patterns. The expected increase in intensity and frequency of precipitation events can intensify urban flooding and sewer backup in the future. Urban drainage systems are the first line of defence against pluvial flooding. However, their design has been widely based on the exceedance probability of historical rainfall events. The underlying assumption that the probability distribution of precipitation extremes is statistically stationary is invalid. Therefore, municipalities need to investigate the potential impact of climate change and plan appropriate adaptation alternatives. In this chapter, we presented a generic framework for climate change impact and adaptation planning of stormwater drainage infrastructure. Important steps in performance assessment, hazard, and risk quantification are discussed. The impacts of climate change on the stormwater system is assessed using 1D-2D simulations for present conditions as well as three climate projection scenarios. There are a wide range of adaptation measures available to reduce the expected impact of climate change, including SUDS—also referred to as BMP or LID. SUDS are gaining wide acceptance due to their natural approach to managing stormwater and effectiveness in improving water pollution and reducing flood susceptibility. Other structural adaptation options, like constructing underground storage tanks or enlargement of pipe size are also viable alternatives. The decision regarding the type of adaptation measure to implement should be based on economic analysis of all feasible options.

3.5 Potential Design and Practice Evolution

Climate change is a big challenge for municipalities across Canada and adapting stormwater systems to cope with the increasing precipitation is crucial to building a resilient community. Effective adaptation planning will require detail investigation on the magnitude and location of the expected impacts. Furthermore, accurate information on the cost and performance of adaptation alternatives like green infrastructure is needed. However, the process of incorporating climate change into existing planning processes can be a difficult task for local decision-makers. Municipalities have a constrained budget and may lack the technical expertise to deal with climate change. Uncertainty in future climate projections and the absence of compressive data on the costs and benefits of adaptation alternatives further complicate the process. For example, the biggest challenge in the case study discussed in this chapter was the lack of accurate and verifiable data to validate the model outputs. Consequently, various assumptions have to be made for the model development and cost-benefit analysis. To overcome such challenges better and more accessible information on the performance of green infrastructure and other adaptation strategies is important. Furthermore, municipal capacity building and training on the full value of green infrastructure and how to integrate it into other projects, such as highway improvements, infrastructure rehabilitation, and urban redevelopment programs is crucial.

Considering future climate conditions in the planning and management process of urban drainage systems is important. Most design guidelines currently stipulate the use of existing IDF curves that are prepared from historical rainfall data. It will be difficult to completely base the planning procedures on predicted future precipitation patterns due to the uncertainties in climate modelling. However, both historical and update IDF curves can be used together to make more robust decisions. For new development sites, municipalities should consider climate change impact assessment and adaptation options as one part of the plan approval process.

Stormwater infrastructure designs in the face of climate change can generally follow three main principles. They can be designed using historic flows, without any adjustment to design flows to consider climate change and accept a gradual decrease in the LOS over the planning horizon. This approach can be acceptable when the expected flood damage is minor. On the other hand, the design can be based on predicted future flows, adjusted from climate change projection scenarios. This can provide enough hydraulic capacity throughout the design life of the infrastructure, however, it could require high sunk cost and financial burden to the developer early on. Another alternative is to design for the present climate with sufficient flexibility to allow the incorporation of additional capacity in the future. Planning for a shorter service life and retrofit when conditions necessitate provides better flexibility to manage uncertainty. However, upgrade costs can still be substantial. Therefore, using a combination of the three approaches for different drainage components can provide optimal results.

3.6 Open Research Questions

Further work in the area of stormwater drainage system adaptation should focus on:

- Performance assessment of adaptation measures and BMPs
- Economic evaluation and cobenefits of adaptation measures like green roofs, pervious pavements, and other infiltration systems
- Improvement in flood loss estimation and fragility curves
- Downscaling of climate data and updating IDF curves at the urban catchment levels
- Risk of storm systems failure considering both structural deterioration and hydraulic capacity reduction due to climate change
- Stormwater infrastructure (pipes, manholes, etc.) deterioration modelling considering climate variables and other environmental factors
- Climate change impact on operation and maintenance of stormwater infrastructure such as storm drain flushing/cleaning frequency
- Design optimisation of drainage systems taking into account specific environmental conditions and potential climate change scenarios

Acknowledgements

The financial support through MITACS and Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant Program (RGPIN-2014-05013) is acknowledged.

References

Abebe, Y., Kabir, G., Tesfamariam, S., 2018. Assessing urban areas vulnerability to pluvial flooding using GIS applications and Bayesian belief network model. J. Clean. Prod., Elsevier 174, 1629–1641.

Arnbjerg-Nielsen, K., Fleischer, H., 2009. Feasible adaptation strategies for increased risk of flooding in cities due to climate change. Water Sci. Technol. 2 (60), 273–281.

Bailey, M., 1992. Regulations, standards and specifications for the design and installation of drainage systems. https://www.vernon.ca/sites/default/files/docs/bylaws/schedule_f.pdf%3e. (Oct. 20, 2017).

Bastidas-Arteaga, E., Stewart, M.G., 2015. Damage risks and economic assessment of climate adaptation strategies for design of new concrete structures subject to chloride-induced corrosion. Struct. Saf., Elsevier 52, 40–53.

Berggren, K., Olofsson, M., Viklander, M., Svensson, G., Gustafsson, A.-M., 2011. Hydraulic impacts on urban drainage systems due to changes in rainfall caused by climatic change. J. Hydrol. Eng. 17 (1), 92–98.

Canadian Infrastructure Report Card, 2016. Stormwater Infrastructure. http://www.canadainfrastructure.ca/en/%3e. (Oct. 12, 2017).

Eden, J.M., Widmann, M., Maraun, D., Vrac, M., 2014. Comparison of GCM- and RCM-simulated precipitation following stochastic postprocessing. J. Geophys. Res. Atmos. 119 (19), 11,040–11,053.

Environment and Climate Change Canada, 2016. Canadian Environmental Sustainability Indicators. https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/temperature-change.html%3e. (Oct. 30, 2017).

Environment and Climate Change Canada, 2017. Engineering Climate Datasets. http://climate.weather.gc.ca/prods_servs/engineering_e.html%3e. (Apr. 19, 2017).

Groisman, P.Y., Easterling, D.R., 1993. Variability and trends of total precipitation and snowfall over the united state and Canada. J. Climate 7 (1), 184–204.

Guidolin, M., et al., 2016. A weighted cellular automata 2D inundation model for rapid flood analysis. Environ. Model. Softw. 84, 378–394.

Hanssen-Bauer, I., Achberger, C., Benestad, R., Chen, D., Førland, E., 2005. Statistical downscaling of climate scenarios over Scandinavia. Climate Res. 29 (3), 255–268.

Harbor, J.M., 1994. A practical method for estimating the impact of land-use change on surface runoff, groundwater recharge and wetland hydrology. J. Am. Plan. Assoc., Taylor & Francis 60 (1), 95–108.

Jacobeit, J., Hertig, E., Seubert, S., Lutz, K., 2014. Statistical downscaling for climate change projections in the Mediterranean region: methods and results. Reg. Environ. Change, Springer 14 (5), 1891–1906.

Kang, N., Kim, S., Kim, Y., Noh, H., Hong, S., Kim, H., 2016. Urban drainage system improvement for climate change adaptation. Water 8 (7), 268.

Kirshen, P., Caputo, L., Vogel, R., Mathisen, P., Rosner, A., Renaud, T., 2014. Adapting urban infrastructure to climate change: a drainage case study. J. Water Resour. Plan. Manag. 141 (4), 1–11.

Kubal, C., Haase, D., Meyer, V., Scheuer, S., 2009. Integrated urban flood risk assessment—adapting a multicriteria approach to a city. Nat. Hazards Earth Syst. Sci. 9, 1881–1895.

Lee, H., McIntyre, N., Wheater, H., Young, A., 2005. Selection of conceptual models for regionalisation of the rainfall-runoff relationship. J. Hydrol. 312 (1–4), 125–147.

Lekuthai, A., Vongvisessomjai, S., 2001. Intangible flood damage quantification. Water Resour. Manag. 15, 343–362.

Lhomme, J., Sayers, P.B., Gouldby, B.P., Samuels, P.G., Wills, M., Mulet-Marti, J., 2008. Recent development and application of a rapid flood spreading method. In: Food Risk Management: Research and Practice, Oxford, UK.

Liu, Y., Pender, G., 2010. A new rapid flood inundation model. In: Proceedings of the first IAHR European Congress, 4–6.

Maksimović, Č., Prodanović, D., Boonya-Aroonnet, S., Leitão, J.P., Djordjević, S., Allitt, R., Allit, R., 2009. Overland flow and pathway analysis for modelling of urban pluvial flooding. J. Hydraul. Res. 47 (4), 512–523.

Merz, B., Kreibich, H., Schwarze, R., Thieken, A., 2010. Assessment of economic flood damage. Nat. Hazards Earth Syst. Sci. 10 (8), 1697–1724.

Mirhosseini, G., Srivastava, P., Stefanova, L., 2013. The impact of climate change on rainfall intensity-duration-frequency (IDF) curves in Alabama. Reg. Environ. Chang. 13 (1), 25–33.

Nanos, M.G., Filion, Y.R., 2016. Risk-based performance assessment of stormwater drainage networks under climate change: a case study in the city of Kingston. In: World Environmental and Water Resources Congress, pp. 73–81.

Natural Resources Canada, 2017. Canadian Guidelines and Database of Flood Vulnerability Functions. http://hazuscanada.ca/sites/all/files/nrc-canadianguidelines-final_2017-03-30_draft.pdf%3e. (Jun. 22, 2017).

Nguyen, V.-T.-V., Nguyen, T.-D., Cung, A., 2007. A statistical approach to downscaling of sub-daily extreme rainfall processes for climate-related impact studies in urban areas. Water Sci. Technol. Water Supply, IWA Publishing 7 (2), 183.

O'Loughlin, G., Chocat, B., Huber, W., 2010. Rainfall-runoff processes and modelling. J. Hydraul. Res. 34 (2015), 733–751.

Ochoa-Rodríguez, S., Onof, C., Maksimović, Č., Wang, L.-P., Willems, P., Van Assel, J., Gires, A., Ichiba, A., Bruni, G., Ten Veldhuis, M.-C., 2013. Urban Pluvial Flood

Modelling Current Theory and Practice. http://www.raingain.eu/sites/default/files/wp3_review_document.pdf%3e. (Feb. 15, 2017).
Olsen, A., Zhou, Q., Linde, J., Arnbjerg-Nielsen, K., 2015. Comparing methods of calculating expected annual damage in urban pluvial flood risk assessments. Water 7 (1), 255–270.
Pachauri, R.K., Meyer, L., Van Ypersele, J.-P., Brinkman, S., Van Kesteren, L., Leprince-Ringuet, N., Van Boxmeer, F., 2015. Climate Change 2014: Synthesis Report. In: Contribution of Working Groups I, II and III to the fifth assessment report of the Intergovernmental Panel on Climate Change, IPCC.
Pauleit, S., Ennos, R., Golding, Y., 2005. Modelling the environmental impacts of urban land use and land cover change - a study in Merseyside, UK. Landsc. Urban Plan., Elsevier 71 (2–4), 295–310.
Rajurkar, M.P., Kothyari, U.C., Chaube, U.C., 2004. Modelling of the daily rainfall-runoff relationship with artificial neural network. J. Hydrol. 285 (1–4), 96–113.
Rosenberg, E.A., Keys, P.W., Booth, D.B., Hartley, D., Burkey, J., Steinemann, A.C., Lettenmaier, D.P., 2010. Precipitation extremes and the impacts of climate change on stormwater infrastructure in Washington state. Clim. Change 102 (1–2), 319–349.
Sandink, D., Schardong, A., Simonovic, S.P., 2017. Computerized tool for the development of intensity-duration-frequency curves under changing climate—Technical Manual. Available from: https://www.idf-cc-uwo.ca/Manuals. (accessed 18 November 2018).
Semadeni-Davies, A., Hernebring, C., Svensson, G., Gustafsson, L.G., 2008. The impacts of climate change and urbanisation on drainage in Helsingborg, Sweden: Combined sewer system. J. Hydrol. 350 (1–2), 100–113.
Simonovic, S.P., Schardong, A., Sandink, D., Srivastav, R., 2016. A web-based tool for the development of intensity duration frequency curves under changing climate. Environ. Model. Softw., Elsevier Ltd 81, 136–153.
Smit, B., et al., 1999. The science of adaptation: a framework for assessment. Mitig. Adapt. Strat. Gl. 4.3-4 199–213.
Srivastav, R.K., Schardong, A., Simonovic, S.P., 2014. Equidistance quantile matching method for updating IDF curves under climate change. Water Resour. Manag. 28 (9), 2539–2562.
Thieken, A.H., Ackermann, V., Elmer, F., Kreibich, H., Kuhlmann, B., Kunert, U., Maiwald, H., Merz, B., Müller, M., Piroth, K., Schwarz, J., Schwarze, R., Seifert, I., Seifert, J., 2009. Methods for the evaluation of direct and indirect flood losses. In: 4th International Symposium on Flood Defence, Toronto, Canada, 6–8.
Van Dijk, E., Van der Meulen, J., Kluck, J., Straatman, J.H.M., 2014. Comparing modelling techniques for analysing urban pluvial flooding. Water Sci. Technol. 69 (2), 305–311.
Wang, C., Wan, T.R., Palmer, I.J., 2010. Urban flood risk analysis for determining optimal flood protection levels based on digital terrain model and flood spreading model. Visual Comput., Springer-Verlag 26 (11), 1369–1381.
Waters, D., Watt, E.W., Marsalek, J., Anderson, B.C., 2003. Adaptation of a storm drainage system to accommodate increased rainfall resulting from climate change. J. Environ. Plan. Manag. 46 (5), 755–770.
Wilby, R.L., Wigley, T.M.L., Conway, D., Jones, P.D., Hewitson, B.C., Main, J., Wilks, D.S., 1998. Statistical downscaling of general circulation model output: a comparison of methods. Water Resour. Res. 34 (11), 2995–3008.
Zhou, Q., 2014. A review of sustainable urban drainage systems considering the climate change and urbanization impacts. Water 6 (4), 976–992.
Zhou, Q., Arnbjerg-Nielsen, K., Mikkelsen, P.S., Nielsen, S.B., Halsnæs, K., 2011. Urban drainage design and climate change adaptation decision making (Ph.D. dissertation). DTU Environment.
Zhou, Q., Halsnaes, K., Arnbjerg-Nielsen, K., 2012. Economic assessment of climate adaptation options for urban drainage design in Odense, Denmark. Water Sci. Technol. 66 (8), 1812–1820.